

Comelia Soltanirad

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Education

Case Western Reserve University, School of Medicine, Cleveland, OH
Master of Science in Biostatistics — Expected Graduation: 2026

The Ohio State University, College of Arts & Sciences, Columbus, OH
Bachelor of Arts in Chemistry — Graduation: May 2025

Honors, Awards & Scholarships

Dean's List – The Ohio State University (2022)

Deputy Director of the Year, Sustainability – Undergraduate Student Government (2023)

Publications

Clark, Ted M., et al. "Comparing the performance of college chemistry students with ChatGPT for calculations involving acids and bases." *Journal of Chemical Education*.

Presentations

Comparing the Performance of Students with ChatGPT for Calculations Involving Acids and Bases. Ohio PKAL Regional Spring Meeting, Kenyon College, Gambier, Ohio. 5/13/2023.

A Literature Review of Student Sense of Belonging in STEM Courses. Ohio PKAL Regional Spring Meeting, Otterbein University, Westerville, Ohio. 4/6/2024.

Research Experience

Research Assistant | Mihaylova Lab, Comprehensive Cancer Center, Columbus, OH
10/2021–05/2023

- Harvested cells for RNA and protein extraction for downstream RT-qPCR and Western blot analysis.
- Generated overexpression vectors through molecular cloning techniques.
- Performed immunofluorescent and immunohistochemical staining of mouse tissue samples.
- Maintained tissue culture lines regularly.
- Assisted the mouse colony manager with routine genotyping.

Research Assistant | Clark Lab, Department of Chemistry and Biochemistry, Columbus, OH
05/2023 – 08/2025

- Utilized AI tools (ChatGPT) to generate responses for chemistry assessments and compared student performance to model outputs.
- Contributed to research examining AI-enhanced learning and its pedagogical implications in chemistry education.
- Gained expertise in applying generative AI technologies to instructional settings.

Teaching Experience

Teaching Assistant – ENGR 399: Engineering Projects, Case Western Reserve University, Cleveland, OH
08/2024–12/2025

- Assist in grading technical reports, project summaries, and data-driven engineering assignments.
- Provide feedback on teamwork, design decisions, and project methodology.
- Support student groups in structuring analyses, organizing deliverables, and applying engineering concepts.

- Communicate regularly with the course instructor to ensure consistent grading and expectations.

Field Experience

US Lab Tech Support Intern | METTLER-TOLEDO, Columbus, OH

05/2023–05/2024

- Assisted in running customer methods including Karl Fischer Titrations and thermal analyses (DSC, TGA).
- Ensured quality assurance of laboratory equipment by testing balances and verifying pH probes for accurate data acquisition.
- Maintained a clean and organized laboratory environment.

Work Experience

Personal Trainer | Recreation & Physical Activity Center (RPAC), The Ohio State University, Columbus, OH

01/2024–08/2025

- Conducted fitness assessments to understand clients' fitness levels and goals.
- Designed and implemented individualized workout programs tailored to client needs.
- Provided guidance on nutrition, lifestyle, and exercise techniques for optimal outcomes.

Barista | Kittie's Cakes, Columbus, OH

06/2022–11/2024

- Mastered espresso preparation and fulfilled 100+ daily customer orders.
- Demonstrated comprehensive menu knowledge and offered personalized drink recommendations.
- Collaborated with baristas, staff, and managers to streamline operations.

Barista | Rohr Café at The Ohio State University, Columbus, OH

08/2023–05/2024

- Operated espresso machines, blenders, and beverage equipment with precision.
- Maintained cleanliness and organization of the café, including dining and prep areas.

Campus Involvement

Director, Sustainability Committee, Undergraduate Student Government, The Ohio State University

05/2024–05/2025

- Engaged faculty to organize sustainability-focused events across campus.
- Managed a team of 6 deputy directors and 20 committee representatives.

Deputy Director, Sustainability Committee, Undergraduate Student Government, The Ohio State University

05/2022–05/2024

- Collaborated with six local businesses to host OSU's first Sustainable Fashion Show.
- Delivered lectures on sustainable fashion and conscious consumption.
- Organized two clothing exchanges to promote slow fashion.
- Planned a sustainable yoga class for 35 students to advocate sustainable wellness.

Committee Representative, Sustainability Committee, Undergraduate Student Government, The Ohio State University

09/2021–05/2022

- Organized a campus-wide clothing exchange, coordinating collection, sorting, and outreach.
- Designed surveys to gather student feedback about dormitory sustainability and living conditions.

Director of Finance | Scarlett Magazine, The Ohio State University

05/2023–05/2024

- Developed semester budgets and optimized resource allocation.
- Aligned organizational finances with executive board goals and initiatives.

Skills

Statistical Programming & Data Science

R: data wrangling (dplyr, tidyr), exploratory data analysis, ggplot2; linear & logistic regression, diagnostics; NHANES data; confidence intervals; bootstrapping; Quarto.

SAS: DATA step programming; PROC SQL, MEANS, FREQ, TRANSPOSE; PROC IMPORT; LIBNAME XLSX; ODS.

Python: pandas, numpy, matplotlib, scikit-learn; data cleaning; visualization; machine learning workflows.

Laboratory Techniques

RT-qPCR · Western Blotting · Molecular Cloning · Tissue Culture · Immunofluorescence · Immunohistochemistry · Quality Control (DSC, TGA)

Other Skills

Scientific Writing · Public Speaking · Leadership & Team Management · Project Coordination

Certifications

ACE Certified Personal Trainer (CPT) – American Council on Exercise (10/2023–Present)

CPR Certification – American Red Cross, Columbus, Ohio (11/2025)